

Section 1 — Project Introduction & Load Dataset

AtliQ Motors EV Market Analysis

Introduction

AtliQ Motors is a leading Electric Vehicle (EV) manufacturer based in the USA and is planning to expand into the Indian market.

The main objective of this project is to analyze the Indian EV market using Python. The analysis includes Exploratory Data Analysis (EDA), Data Cleaning, and Feature Engineering to prepare a clean and reliable dataset.

The final cleaned dataset will be used to build an interactive Power BI dashboard and generate meaningful business insights that can support better decision-making.

```
In [3]: # Import required libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset

df = pd.read_excel(r"C:\Users\Pranav\EV_Project\Datasets\AtliQ_Ev_raw.xlsx")

# Display the first 5 rows

df.head()
```

Out[3]:

	Date	Fiscal_year	Quarter	Vehicle_category	Maker	Electric_vehicles_sold_maker
0	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0
1	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0
2	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0
3	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0
4	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0

Observation

- The dataset has been loaded successfully.
- It contains information about EV sales by different manufacturers and states in India.
- Before cleaning the data, we will first inspect the dataset to understand its structure and identify any data quality issues.

Section 2 — Dataset Overview

```
In [6]: # Check the size of the dataset
print("Dataset Shape:")
print(df.shape)

# Display all column names
print("\nColumn Names:")
print(df.columns)

# Check data types and missing values
print("\nDataset Information:")
df.info()

# Summary statistics for numerical columns
print("\nSummary Statistics:")
display(df.describe())
```

Dataset Shape:
(55398, 9)

Column Names:

```
Index(['Date', 'Fiscal_year', 'Quarter', 'Vehicle_category', 'Maker',
      'Electric_vehicles_sold_maker', 'State',
      'Electric_vehicles_sold_bystates', 'Total_vehicles_sold'],
      dtype='object')
```

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 55398 entries, 0 to 55397

Data columns (total 9 columns):

#	Column	Non-Null	Count	Dtype
0	Date	55398	non-null	datetime64[ns]
1	Fiscal_year	55398	non-null	int64
2	Quarter	55398	non-null	object
3	Vehicle_category	55398	non-null	object
4	Maker	55398	non-null	object
5	Electric_vehicles_sold_maker	55398	non-null	int64
6	State	55398	non-null	object
7	Electric_vehicles_sold_bystates	55392	non-null	float64
8	Total vehicles sold	55398	non-null	int64

```
dtypes: datetime64[ns](1), float64(1), int64(3), object(4)
```

```
memory usage: 3.8+ MB
```

Summary Statistics:

	Date	Fiscal_year	Electric_vehicles_sold_maker	Electric_vehicles_sold
count	55398	55398.000000	55398.000000	55398
mean	2022-09-20 23:29:38.360229632	2023.014694	2529.969620	85
min	2021-04-01 00:00:00	2022.000000	0.000000	
25%	2022-01-01 00:00:00	2022.000000	42.000000	
50%	2022-10-01 00:00:00	2023.000000	663.000000	5
75%	2023-07-01 00:00:00	2024.000000	2633.000000	53
max	2024-03-01 00:00:00	2024.000000	44630.000000	2666
std	NaN	0.813026	4760.231732	219



Observation

- The dataset overview helps us understand its overall structure.
- We checked the total number of rows and columns.
- The column names were reviewed to understand the available information.

- The first few records were displayed to get a quick look at the data.
- The dataset information shows the data type of each column and the number of non-null values.
- Summary statistics provide an overview of the numerical columns.
- In the next section, we will inspect missing values in the dataset.

Section 3 — Missing Values Analysis

```
In [7]: # Check missing values in each column
print("Missing Values in Each Column:")
print(df.isnull().sum())

# Calculate missing values in percentage
missing_percentage = (df.isnull().sum() / len(df)) * 100
print("\nMissing Values Percentage:")
print(missing_percentage.round(2))
```

Missing Values in Each Column:

Date	0
Fiscal_year	0
Quarter	0
Vehicle_category	0
Maker	0
Electric_vehicles_sold_maker	0
State	0
Electric_vehicles_sold_bystates	6
Total_vehicles_sold	0

dtype: int64

Missing Values Percentage:

Date	0.00
Fiscal_year	0.00
Quarter	0.00
Vehicle_category	0.00
Maker	0.00
Electric_vehicles_sold_maker	0.00
State	0.00
Electric_vehicles_sold_bystates	0.01
Total_vehicles_sold	0.00

dtype: float64

Observation

- We checked the dataset for missing values in each column.
- The number and percentage of missing values help us identify which columns need attention.
- Based on the inspection, the `electric_vehicles_sold_bystates` column contains missing (`#N/A`) values.
- These values will be handled during the Data Cleaning phase.

Section 4 — Duplicate Records Analysis

```
In [8]: # Check duplicate records
duplicate_rows = df.duplicated().sum()
print("Total Duplicate Rows:", duplicate_rows)

# Display duplicate records (if any)
if duplicate_rows > 0:
    print("\nDuplicate Records:")
    display(df[df.duplicated()])
else:
    print("\nNo duplicate records found.")
```

Total Duplicate Rows: 0

No duplicate records found.

Observation

- We checked the dataset for duplicate records.
- Duplicate rows can lead to incorrect analysis and inaccurate results.
- If duplicate records are found, they will be removed during the Data Cleaning phase.
- If no duplicates are found, no action is required.

Section 5 — Data Quality Check

```
In [10]: # Check data types of all columns
print("Data Types:")
print(df.dtypes)

# Check unique values in the Maker column
print("\nUnique Maker Names:")
print(df["Maker"].unique())

# Check for extra spaces in the Maker column
extra_spaces = df["Maker"] != df["Maker"].str.strip()
print("\nRows with Extra Spaces in Maker Column:")
print(extra_spaces.sum())

# Display rows having extra spaces
display(df[extra_spaces])
```

```
Data Types:
Date                               datetime64[ns]
Fiscal_year                       int64
Quarter                           object
Vehicle_category                  object
Maker                             object
Electric_vehicles_sold_maker      int64
State                             object
Electric_vehicles_sold_bystates   float64
Total_vehicles_sold               int64
dtype: object
```

Unique Maker Names:

```
['OLA ELECTRIC' ' OLA ELECTRIC' 'BYD India' ' BYD India'
 'PCA Automobiles' ' PCA Automobiles' 'BMW India' 'Volvo Auto India'
 ' Volvo Auto India' 'KIA Motors' 'Tata Motors' 'MG Motor' ' MG Motor'
 'Mahindra & Mahindra' 'Hyundai Motor' 'Mercedes -Benz AG'
 ' Mercedes -Benz AG' 'HERO ELECTRIC' 'OKINAWA' 'AMPERE' 'ATHER'
 'PURE EV' 'TVS' 'REVOLT' 'BAJAJ' 'BEING' 'JITENDRA' 'OTHERS' 'OKAYA EV'
 'BGAUSS' 'BATTRE ELECTRIC' 'KINETIC GREEN']
```

Rows with Extra Spaces in Maker Column:

8

	Date	Fiscal_year	Quarter	Vehicle_category	Maker	Electric_vehicles_sold_mak
9	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	
28	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	
55	2021-04-01	2022	Q1	4-Wheelers	BYD India	
106	2021-04-01	2022	Q1	4-Wheelers	PCA Automobiles	
111	2021-04-01	2022	Q1	4-Wheelers	PCA Automobiles	
187	2021-04-01	2022	Q1	4-Wheelers	Volvo Auto India	
311	2021-04-01	2022	Q1	4-Wheelers	MG Motor	1
440	2021-04-01	2022	Q1	4-Wheelers	Mercedes - Benz AG	

Observation

- We reviewed the data types of all columns.
- The unique values in the **Maker** column were inspected to identify any inconsistent text formatting.
- Some manufacturer names contain leading or trailing spaces.
- These extra spaces can create duplicate categories during analysis.

- The **Maker** column will be cleaned by removing unnecessary spaces in the Data Cleaning section.

Section 6 — Data Cleaning

```
In [11]: # Fill missing values with 0
df["Electric_vehicles_sold_bystates"] = df["Electric_vehicles_sold_bystates"].fillna(0)

# Remove extra spaces from the Maker column
df["Maker"] = df["Maker"].str.strip()

# Convert the column to integer
df["Electric_vehicles_sold_bystates"] = df["Electric_vehicles_sold_bystates"].astype(int)

# Reset the index
df.reset_index(drop=True, inplace=True)

# Check missing values after cleaning
print("Missing Values After Cleaning:")
print(df.isnull().sum())

# Display first 5 rows after cleaning
display(df.head())
```

Missing Values After Cleaning:

Date	0
Fiscal_year	0
Quarter	0
Vehicle_category	0
Maker	0
Electric_vehicles_sold_maker	0
State	0
Electric_vehicles_sold_bystates	0
Total_vehicles_sold	0
dtype:	int64

	Date	Fiscal_year	Quarter	Vehicle_category	Maker	Electric_vehicles_sold_maker	
0	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	
1	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	
2	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	A
3	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	Al
4	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	Al

Observation

- Missing values in the **Electric_vehicles_sold_bystates** column have been replaced with **0**.
- Extra spaces were removed from the **Maker** column to ensure consistent manufacturer names.
- The **Electric_vehicles_sold_bystates** column was converted to an integer data type after handling missing values.
- The dataset index was reset to keep the data organized.
- No duplicate records were removed because no duplicate rows were found during the data inspection.
- The dataset is now clean and ready for feature engineering.

Section 7 — Feature Engineering

```
In [12]: # Convert Date column to datetime format
df["Date"] = pd.to_datetime(df["Date"])

# Create Year column
df["Year"] = df["Date"].dt.year

# Create Month column
df["Month"] = df["Date"].dt.month_name()

# Create Month Number column
df["Month_Number"] = df["Date"].dt.month

# Create EV Penetration (%) column
df["EV_Penetration_%"] = (
```



```

df["Electric_vehicles_sold_bystates"] /
df["Total_vehicles_sold"]
) * 100
df["EV_Penetration_%"] = df["EV_Penetration_%"].round(2)

# Create EV Penetration Category column
df["EV_Penetration_Category"] = "Low"
df.loc[df["EV_Penetration_%"] >= 5, "EV_Penetration_Category"] = "Medium"
df.loc[df["EV_Penetration_%"] >= 10, "EV_Penetration_Category"] = "High"

# Display first 5 rows
display(df.head())

```

	Date	Fiscal_year	Quarter	Vehicle_category	Maker	Electric_vehicles_sold_maker	
0	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	
1	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	
2	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	A
3	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	Al
4	2021-04-01	2022	Q1	2-Wheelers	OLA ELECTRIC	0	Al

Observation

- The **Date** column was converted into datetime format for easier date-based analysis.
- Three new columns (**Year**, **Month**, and **Month_Number**) were created from the Date column.
- The **EV_Penetration_%** column was calculated using the ratio of electric vehicles sold to total vehicles sold and rounded to two decimal places.
- A new **EV_Penetration_Category** column was created to classify EV adoption into **Low**, **Medium**, and **High** categories.
- The dataset now contains additional business-friendly features that will be useful for Power BI dashboards and further analysis.

Section 8 — Final Dataset & Export

```

In [13]: # Arrange the columns
df = df[
    [
        "Date",
        "Fiscal_year",

```

```

        "Quarter",
        "Year",
        "Month",
        "Month_Number",
        "State",
        "Maker",
        "Vehicle_category",
        "Electric_vehicles_sold_maker",
        "Electric_vehicles_sold_bystates",
        "Total_vehicles_sold",
        "EV_Penetration_%",
        "EV_Penetration_Category"
    ]

# Display the final dataset
print("Final Dataset:")
display(df.head())

# Export the cleaned dataset
output_path = r"C:\Users\Pranav\EV_Project\Datasets\Atliq_ev_market_clean.xlsx"

df.to_excel(output_path, index=False)

print("Dataset exported successfully.")
print("Location:", output_path)

```

Final Dataset:

	Date	Fiscal_year	Quarter	Year	Month	Month_Number	State	Maker	Vehicle
0	2021-04-01	2022	Q1	2021	April	4	Sikkim	OLA	ELECTRIC
1	2021-04-01	2022	Q1	2021	April	4	Sikkim	OLA	ELECTRIC
2	2021-04-01	2022	Q1	2021	April	4	Andaman & Nicobar Island	OLA	ELECTRIC
3	2021-04-01	2022	Q1	2021	April	4	Arunachal Pradesh	OLA	ELECTRIC
4	2021-04-01	2022	Q1	2021	April	4	Arunachal Pradesh	OLA	ELECTRIC

Dataset exported successfully.

Location: C:\Users\Pranav\EV_Project\Datasets\Atliq_ev_market_clean.xlsx

Observation

- The final dataset was organized in a clear and consistent column order.
- All cleaned and newly created feature engineering columns were included.
- The cleaned dataset was exported as **Atliq_ev_market_clean.xlsx**.

- The dataset is now ready for Power BI dashboard development and business analysis.

Conclusion

This project focused on preparing a clean and business-ready dataset for analyzing the Indian Electric Vehicle (EV) market.

During the analysis, the dataset was carefully inspected to identify data quality issues before performing any cleaning. Missing values in the **Electric_vehicles_sold_bystates** column were replaced with **0**, and extra spaces in the **Maker** column were removed to ensure consistency. No duplicate records were found in the dataset.

Feature engineering was performed by creating the **Year**, **Month**, **Month_Number**, **EV_Penetration_%**, and **EV_Penetration_Category** columns, making the dataset more suitable for analysis.

Finally, the cleaned dataset was exported as **Atliq_ev_market_clean.xlsx** and is now ready for building an interactive Power BI dashboard and generating meaningful business insights for AtliQ Motors.

In []: